

Exercise: Monitor Tropical Cyclone Biparjoy

- Biparjoy: June 6-16, 2023
- Region: Arabian Sea
- Wind speed: max. 194 km/h
- Diameter: max. 567 km
- Air pressure: below 945 mbar
- Saffir-Simpson scale: category 3
- **Affected regions**
Western India



Exercise: Tropical Cyclone Biparjoy

This is a two-part exercise using [Giovanni](#):

- Part 1: Access and analysis of GPM-IMERG precipitation using Giovanni
- Part 2: Access and Analysis of sea level pressure and winds from MERRA-2 using Giovanni



Part 1: GPM-IMERG Data Access and Analysis

1. Go to Giovanni: <http://giovanni.gsfc.nasa.gov/giovanni>
2. On the Giovanni page you will see the following options:
 - **Select Plot:** allows selection of analysis options
 - **Select Data Range:** allows selection of a time period
 - **Select Region (Bounding Box or Shape):** allows selection of a geographic region by latitude-longitude, map, or shapefile
 - **Keyword:** allows search of data parameters by keyword
 - **Plot Data:** (located on the bottom right of the page) begins the action to make a desired plot

Part 1: IMERG Rainfall Maps

3. For **Keyword**: Enter **IMERG v07 Final**

– Select:

☒	Multi-satellite precipitation estimate with gauge calibration - Final Run (recommended for general use) (GPM_3IMERGHH v07)	mm/hr ▾	GPM	Half-Hourly
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– Under **Select Plot** Select: **Animation**

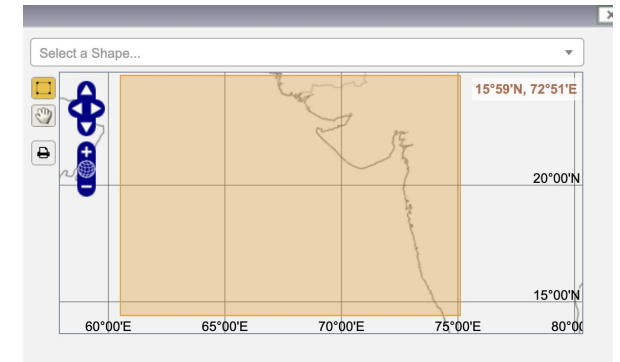
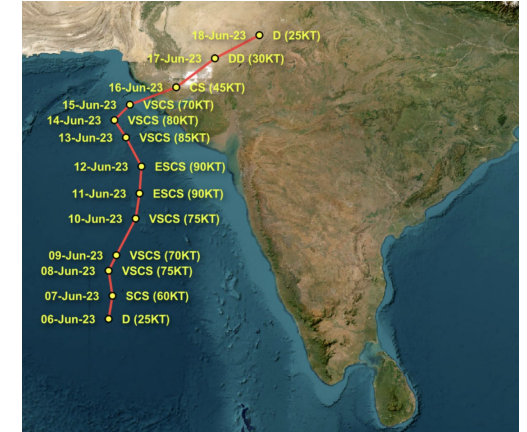
– Under Select Date Range (UTC)

- In the **YYYY-MM-DD** windows,
enter **2023-06-14** to start and

2023-06-16 for the end date

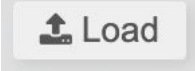
– Click on **Plot Data** (on the bottom right of the screen)

– You will get an animation of half-hourly rainfall



Part 2: MERRA-2 Sea Level Pressure and Surface Wind Speed

3. Click on **History** ☐ **1. Animation** ☐ **User Input** (Left Menu Bar)

- Select 
- **For Keyword:** Enter MERRA-2 Hourly, you will see a list of parameters
- Select **Sea level pressure time average**

☒	<u>Sea level pressure, time average</u> (M2T1NXSLV v5.12.4)	hPa	MERRA-2 Reanalysis	Hourly
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- Keep other options the same
- Click on **Plot Data** (on the bottom right of the screen)
- A workflow will start and you will get an animation of Sea Level Pressure for the cyclone.

4. Repeat the above steps but select the following while uncheck the Sea level pressure

☒	<u>Surface wind speed, time average</u> (M2T1NXFLX v5.12.4)	m s ⁻¹	MERRA-2 Reanalysis	Hourly
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